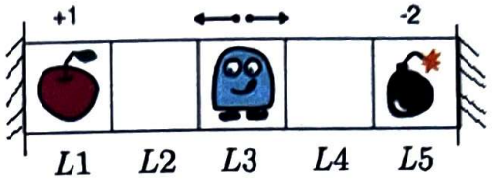
Markov Decision Processes

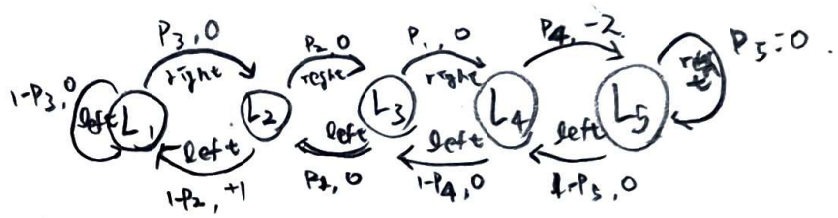
(Q1) 1x5 Grid World

- State : $S = \{L_1, \cdots, L_5\}$
- Reward: 사과 (+1), 폭탄 (-2), 빈칸 (0), 벽(0)



⇒ Draw the transition diagram

s	s'	a	$P(s'\|s,a)$	$r(s,a,s')$
L_3	L_2	left	$1-P_1$	0
L_3	L_4	right	P_1	0
L_2	L_1	left	$1-P_2$	+1
L_2	L_3	right	P_2	0
L_1	L_1	left	$1-P_3$	0
L_1	L_2	right	P_3	0
L_4	L_3	left	$1-P_4$	0
L_4	L_5	right	P_4	-2
L_5	L_5	right	P_5	0
L_5	L_4	left	$1-P_5$	0



1x2 Grid World

(Q2) 1x2 Grid World (deterministic case)

$\mu 1$

	L_1	L_2
left	0	0
right	1	1

$\mu 2$

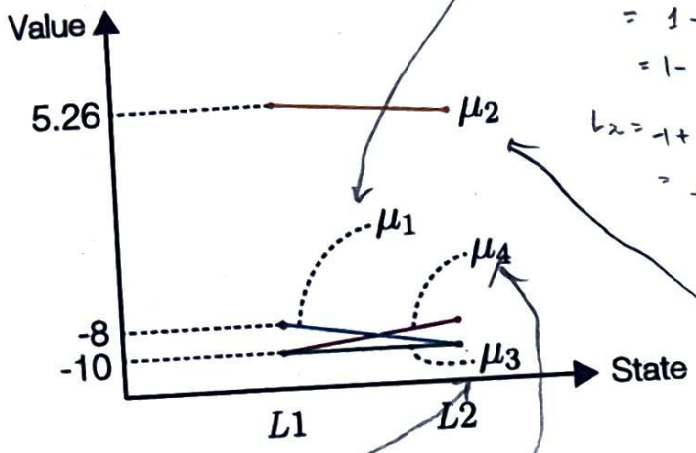
	L_1	L_2
left	0	1
right	1	0

$\mu 3$

	L_1	L_2
left	1	0
right	0	1

$\mu 4$

	L_1	L_2
left	1	1
right	0	0



μ_1

$L_1 = 1 + 0.9(-1) + 0.9^2(-1) \ldots$

$= 1 - 0.9(1 + 0.9 \ldots)$

$= 1 - \frac{0.9}{1-0.9} = -8$

$L_2 = -1 + 0.9(-1) + 0.9^2(-1) \ldots$

$= -1 - \frac{0.9}{1-0.9} = -10$

μ_2

$L_1 = 1 + 0.9 \cdot 0 + (0.9)^2 \times 1 \ldots$

$= 1 + 0.9^2 + 0.9^4 \ldots$

$(0.9^2) = 0.81$

$= \frac{1}{1-0.81} = \frac{1}{0.19} \approx 5.26$

$L_2 = 0 + 0.9 + (0.9)^2 \times 0 \ldots$

$= 0.9 \times 5.26 = 4.75$

Optimal policy = $\mu 2$

μ_3:

$L_1 = -1 + \frac{-1}{1-0.9} + -1 \times 0.9^2$

$= -1 - \frac{0.9}{1-0.9} = -10$

$L_2 = -1 + -1 \times 0.9 \ldots$

$= -10$

μ_4:

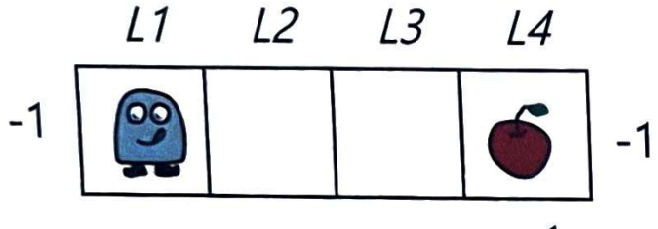
$L_1 = -10$

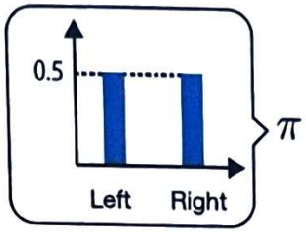
$L_2 = 0 + (0.9)1 + (0.9)^2 1$

$= 0 - \frac{0.9}{1-0.9} = -9$

Quiz

(Q3) 앞 Example 의 2-Grid World 문제를 4-Grid World 문제로 확장하는 경우, state value 에 대한 Bellman equation 을 적고, 각 state value 를 구하라.





$\gamma = 0.9$

$$v_\pi(s) = \sum_a \pi(a|s) \{ r(s,a,s') + \gamma v_\pi(s') \}$$

$$v_\pi(L_1) = 0.5\{-1 + 0.9 v_\pi(L_1)\} + 0.5\{0 + 0.9 v_\pi(L_2)\}$$

$$= 0.5 + 0.45 V_1 + 0.45 V_2$$

$$V_2 = 0.5 \times (0 + 0.9 V_1) + 0.5 \times (0 + 0.9 V_3)$$

$$= 0.45 V_1 + 0.45 V_3$$

$$V_3 = 0.5 \times (0 + 0.9 V_2) + 0.5 (1 + 0.9 V_4)$$

$$= 0.45 V_2 + 0.5 + 0.45 V_4$$

$$V_4 = 0.5 \times (0 + 0.9 V_3) + 0.5 \times (-1 + 0.9 V_4)$$

$$= 0.45 V_3 + 0.45 V_4 - 0.5 \Rightarrow V_4 = \frac{9V_3 - 10}{11}$$

0.11 0.9 0.1

$$0.55 V_1 - 0.45 V_2 = 0.5 \Rightarrow V_1 = \frac{9V_2 + 10}{11}$$

$$0.45 V_1 + 0.45 V_3 = V_2$$

$$-9V_1 + 9V_3 = 20V_2$$

$$-9\left(\frac{9V_2 + 10}{11}\right) + 9V_3 = 20V_2$$

$$-81V_2 + 90 + 220V_2 - 99V_3$$

$$139V_2 - 99V_3 = -90$$

$$-99V_2 + 139V_3 = 2024$$

$$V_3 = \frac{99V_2 + 20}{139}$$

$V_2 = -1.106$

$V_3 = -0.644$

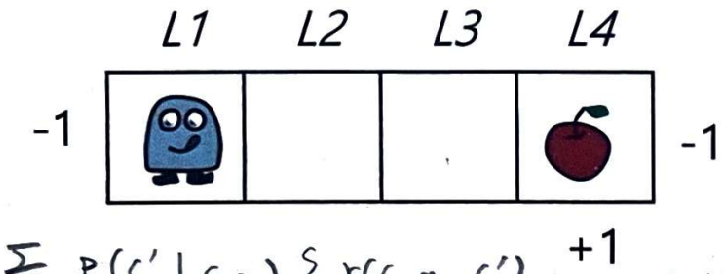
$V_1 = -1.814$

$V_4 = -1.436$

Quiz

(Q4) 앞 Example 의 2-grid World 문제를 확장한 4-Grid World 문제에 대하여, Bellman optimality equation 을 유도하라.

$\gamma = 0.9$



$$V_*(s) = \max_a \sum_{s'} P(s' \mid s,a) \{ r(s,a,s') + \gamma V_*(s') \}$$

L_1

$-1 + 0.9 V_*(L_1)$ $a = \text{Left}$

$0 + 0.9 V_*(L_2)$ $a = \text{Right}$

$V_*(L_1) = \max \{ -1 + 0.9 V_*(L_1), 0.9 V_*(L_2) \}$

$V_1 = 0.9 V_2$

$V_2 = 0.9 V_3$

$V_3 = 1 + 0.9 V_4$

$V_4 = 0.9 V_3$

$V_3 - 0.81 V_3 = 1$

$0.19 V_3 = 1$

$V_3 = \frac{1}{0.19}$

$= 5.263$

$V_4 = 4.737$

$V_2 = 4.737$

$V_1 = 4.263$

L_2

$0 + 0.9 V_*(L_1)$ $a = \text{Left}$

$0 + 0.9 V_*(L_3)$ $a = \text{Right}$

$V_*(L_2) = \max \{ 0.9 V_*(L_1), 0.9 V_*(L_3) \}$

L_3

$0 + 0.9 V_*(L_2)$ $a = \text{left}$

$1 + 0.9 V_*(L_4)$ $a = \text{Right}$

$V_*(L_3) = \max \{ 0.9 V_*(L_2), 1 + 0.9 V_*(L_4) \}$

L_4

$0 + 0.9 V_*(L_3)$ $a = \text{left}$

$-1 + 0.9 V_*(L_4)$ $a = \text{Right}$

$V_*(L_4) = \max \{ 0.9 V_*(L_3), -1 + 0.9 V_*(L_4) \}$